



Long-Term Memory

Long-term memory is a nearly permanent storage space for learned information and experiences. It can hold memories for years. Long-term memory moves short-term memories to this larger space. Your memories may last a long time, but your ability to retrieve them may be more challenging with age, certain conditions or injuries.

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There are two main types of long-term memory that help you retain information you learned.

What Is Long-Term Memory?

Long-term memory is a nearly permanent storage space to hold experiences and information you've learned. There isn't a limit on how much information you can store in your long-term memory. It can stay in this storage space for years.

The main function of long-term memory is to serve as an archive — to make your short-term memory more permanent. Once your short-term memories move to long-term storage, you can retrieve these memories — like the answer to the trivia question on television, the house number of the street you lived on when you were a kid or what you ate for dinner last night.



What are the types of long-term memory?

There are two main types of long-term memory:

- **Declarative memory (explicit memory):** This is the storage of facts, events and locations. You're always adding to this storage space and recalling this information when you need it.
- **Nondeclarative memory (implicit memory):** This is the storage of learned skills and habits or relationships. You aren't consciously aware of when you're learning this information, but you can recall it at any time.

There are two subtypes of declarative memory:

- **Semantic memory:** Learned information about your environment
- **Episodic memory (autobiographical memory):** Learned information about your life events or experiences

There are four subtypes of nondeclarative memory:

- **Procedural memory:** This is the information needed to help you perform tasks. It combines executive skills (like planning and problem-solving) and motor skills (coordination of muscle movements).
- **Priming and perceptual memory:** You can relate something you previously had exposure to in your memory to help you process and learn new information.
- **Associative learning (classical conditioning):** This is linking one thing in your memory with another. It helps you make connections between information you already learned and new

information.

- **Nonassociative learning:** This is adjusting how you respond to something over time. It helps you react to things within your environment.

Examples of the subtypes of nondeclarative memory include:

- **Procedural memory:** You tie your shoes or turn on your phone.
- **Priming and perceptual memory:** You see the color red and select a stop sign, based on your knowledge that the sign shares the same color.
- **Associative learning:** The most common example in history is Pavlov's dog, where a dog salivates when it hears a bell. The dog associated the sound with being fed.
- **Nonassociative learning:** You purchase a new refrigerator and hear it running constantly. Over time, you adjust to the noise and the sound no longer bothers you.

Where is long-term memory stored?

All memories form in your **hippocampus**, which is in your **temporal lobes**. There's a temporal lobe and hippocampus on each side of your head. These parts of your brain sit behind your temples and reach to your ears.

Each type of long-term memory is formed by connecting neurons in various parts of the brain:

- **Associative learning memory:** Amygdala, cerebellum
- **Declarative memory:** Hippocampus
- **Episodic memory:** Hippocampus, temporal lobe, neocortex (cerebral cortex)
- **Nonassociative learning memory:** Reflex pathways (hippocampus)
- **Nondeclarative memory:** Basal ganglia, cerebellum, amygdala
- **Priming and perceptual memory:** Neocortex
- **Procedural memory:** Striatum (forebrain), cerebellum, motor cortex (frontal lobe)
- **Semantic memory:** Temporal cortex (temporal lobe), prefrontal cortex (frontal lobe)



How long does long-term memory last?

There isn't a specific amount of time that long-term memory can last. In many cases, certain memories can last for years, decades even. You may have memories that last a lifetime.

While the memories themselves can last an unlimited amount of time, your ability to retrieve memories from this storage space may not work as well as it used to as you get older. This can happen due to natural changes that happen with age — your brain activity slows down, so it might take you a little longer to find certain memories within your archives.

Some conditions and traumatic events both physical and psychological can cause memory impairment or long-term **memory loss**. If you're having trouble remembering, talk to a healthcare provider.

A note from Cleveland Clinic

Your long-term memory is a very important piece of what makes you who you are. You're able to retrieve information from this part of your brain, often without even thinking about it. It can help you complete routine tasks like brushing your hair or making a cup of coffee in the morning. This database in your brain may make you remember your grandparents' birthdays, how to ride a bicycle or the capital of the state you live in.

Injuries or conditions that impact your brain may put your long-term memory at risk. As you age, natural changes may affect your ability to retrieve memories. A healthcare provider can help you with any memory concerns you might have.

Experts You Can Trust

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References

Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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